

Unidirectional flow (exact)

Velocity field kinematics:

$$\underline{v} = v_x(y) \underline{e}_x \quad (\Rightarrow \underline{v} \cdot \nabla \underline{v} = \underline{0})$$

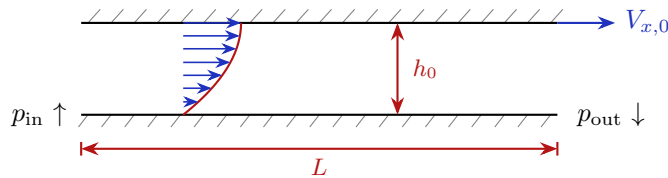
Basic equations:

$$\text{COM:} \quad 0 = 0$$

$$\text{COLM-}x: \quad 0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}$$

$$\text{COLM-}y: \quad 0 = -\frac{\partial p}{\partial y}$$

Typical 2D channel geometry (“combined Poiseuille–Couette flow”):



Many other combinations of BCs possible!

Solution strategy:

1. p is not a function of y (from COLM- y)
2. v_x is not a function of x (from kinematics)
3. each term on RHS of COLM- x must be constant:

$$\frac{\partial p}{\partial x} = \frac{dp}{dx} = \text{const.} = \frac{p_{\text{in}} - p_{\text{out}}}{L} = -\frac{\Delta p}{L}$$

$$v_x(y) = \frac{h_0^2}{2\mu} \left(-\frac{dp}{dx} \right) \left[\frac{y}{h_0} - \left(\frac{y}{h_0} \right)^2 \right] + V_{x,0} \frac{y}{h_0}$$

4. $Q = \int_0^{h_0} v_x(y) dy$ leads to flow rate–pressure drop relation (**done!**)

Lubrication theory (approx)

Velocity field kinematics:

$$\underline{v} = v_x(x, y) \underline{e}_x + v_y(x, y) \underline{e}_y \quad (\text{with } \underline{v} \cdot \nabla \underline{v} \approx \underline{0})$$

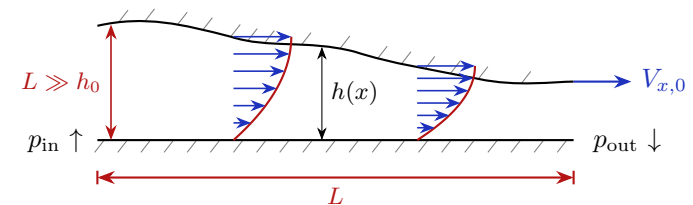
Basic equations:

$$\text{COM:} \quad \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0$$

$$\text{COLM-}x: \quad 0 \approx -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}$$

$$\text{COLM-}y: \quad 0 \approx -\frac{\partial p}{\partial y}$$

Typical 2D channel geometry:



Solution strategy:

1. p is not a function of y (COLM- y)
2. v_x is a function of x ; solve COLM- y for v_x in terms of $\frac{\partial p}{\partial x}$:

$$v_x(x, y) = \frac{h(x)^2}{2\mu} \left(-\frac{\partial p}{\partial x} \right) \left[\frac{y}{h(x)} - \left(\frac{y}{h(x)} \right)^2 \right] + V_{x,0} \frac{y}{h(x)}$$

3. $Q = \int_0^{h(x)} v_x(x, y) dy$ leads to flow rate–pressure gradient (**unknown!**)
4. write COM as $\partial v_y / \partial y = -\partial v_x / \partial x$
 - (a) integrate over *all* y and use BCs to get ODE for $p(x)$
 - (b) integrate up to *arbitrary* y to find $v_y(x, y)$ (optional)